

Fig. 2: Block diagram of a solar system

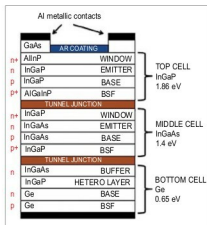


Fig. 3: Structure of a multi-junction solar cell

circuit is not used. The device gets energy from the battery.

Sometimes, a solid copper foundation is laid over the solar panel that gives it massive strength. Solar panels may have anti-reflective (AR) coating over the surface to produce more energy and increase efficiency. AR layer usually has NaOH surface texturation with several tiny pyramids. The layer's thickness should be designed for better efficiency.

Multi-junction structure of a solar cell

A few years ago, late Dr A.P.J. Abdul Kalam said that, we are not utilizing solar energy efficiently. Researchers are working hard and have come up with different solar structures and technologies to increase the efficiency of solar cells. Till now, we have depended on single-junction solar cells that

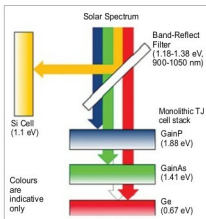


Fig. 4: Concept of a triple-junction cell that targets discrete bands of incoming sunlight using a combination of three layers

give an efficiency of about 20 per cent. To improve this efficiency, the concept of multi-junction solar cells has been adopted. These are solar cells with multiple p-n junctions made up of different semiconductor materials.

The gap between valance band and conduction band is called forbidden gap or energy gap (E_g). In an energy band diagram, electrons jump from valence to conduction band and become free when electrons absorb enough energy. These electrons are considered free electrons and take part in current flow. If an applied photon has wavelength greater than or equal to hc/E_g (h is Planck's constant and c is velocity of light), electrons gain enough energy and jump from valence to conduction band.

Electrons that belong to con-

duction band take part in current flow and are thus responsible for generating electricity. In the above-mentioned relationship, h and c are constant, which means the number of free electrons depends on E_g . This Physics fundamental has been used in the structure of solar cells. A multi-junction structure (material with different values of E_g) absorbs maximum sunlight (broad range of wavelengths), generates more free electrons or current and, ultimately, increases the efficiency of solar cells.

The top-most junction has the widest band gap and, as we go down, it decreases. It is necessary to find the best semiconductor materials in terms of band gap to increase efficiency. Fig. 3 shows the different semiconductor materials used in a multi-junction solar cell. As photons (sunlight) strike the solar cell, highest energy-level photons are absorbed by the top-most layer, and photons with low energy are absorbed in the lower layers.

A window layer is used to reduce surface recombination velocity, and the back surface field (BSF) layer reduces the scattering of carriers towards the tunnel junction. It simply functions as an electrical contact and, thus, covers the entire back surface of the cell structure. It is always made of metal.

Researchers of the University of New South Wales, Australia, have developed a mini-module that combines a silicon cell on one face of a glass prism, with a triple-junction (TJ) solar cell on the other, as shown in Fig. 4. This research has achieved the world record for the highest efficiency, which is approximately 34.5 per cent. This novel model was also tested and confirmed by US National Renewable Energy Laboratory.

The triple-junction cell targets discrete bands of incoming sunlight using a combination of three layers. As sunlight passes through each layer, energy is extracted by each junction.